

Non-Maximum Suppression & Average Precision

Overview

$\text{IoU} = 0.72$

score 0.91 ✓

$\text{AP@.5} = 0.83$

$\text{mAP} = 0.71$

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● The Problem

● NMS

● IoU

● Average Precision

Fundamentals

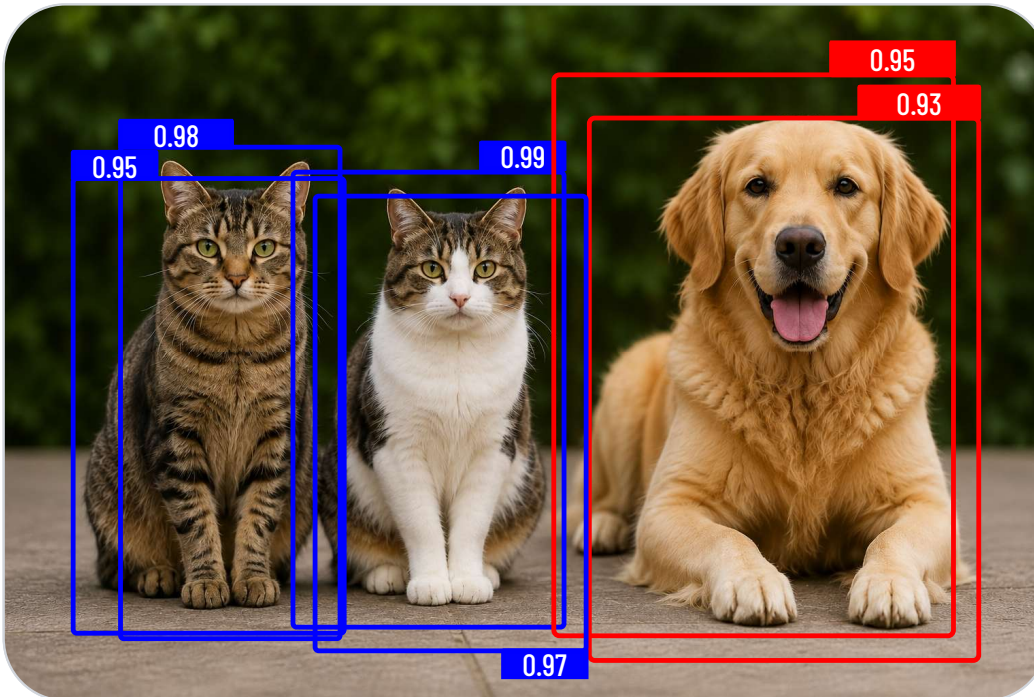


Image generated using ChatGPT. Bounding boxes added manually for demonstration purposes.

Parameter	Rationale
Position + Size	(x, y) In COCO format, this is the top-left corner coordinate plus width and height. In Pascal VOC, it represents the top-left (x_1, y_1) and bottom-right (x_2, y_2) corners.
Class	The class label as predicted by the object detector. <i>Example: Dog, Cat</i>
Confidence Score	The probability that the bounding box contains the object as indicated by the class label. Value between 0 and 1.

Detectors are enthusiastic

⚠ The Problem

Object detectors such as **YOLO**, **Faster R-CNN** predict hundreds of candidate bounding boxes per image. For each detected object, **multiple overlapping boxes** are produced with different confidence scores.

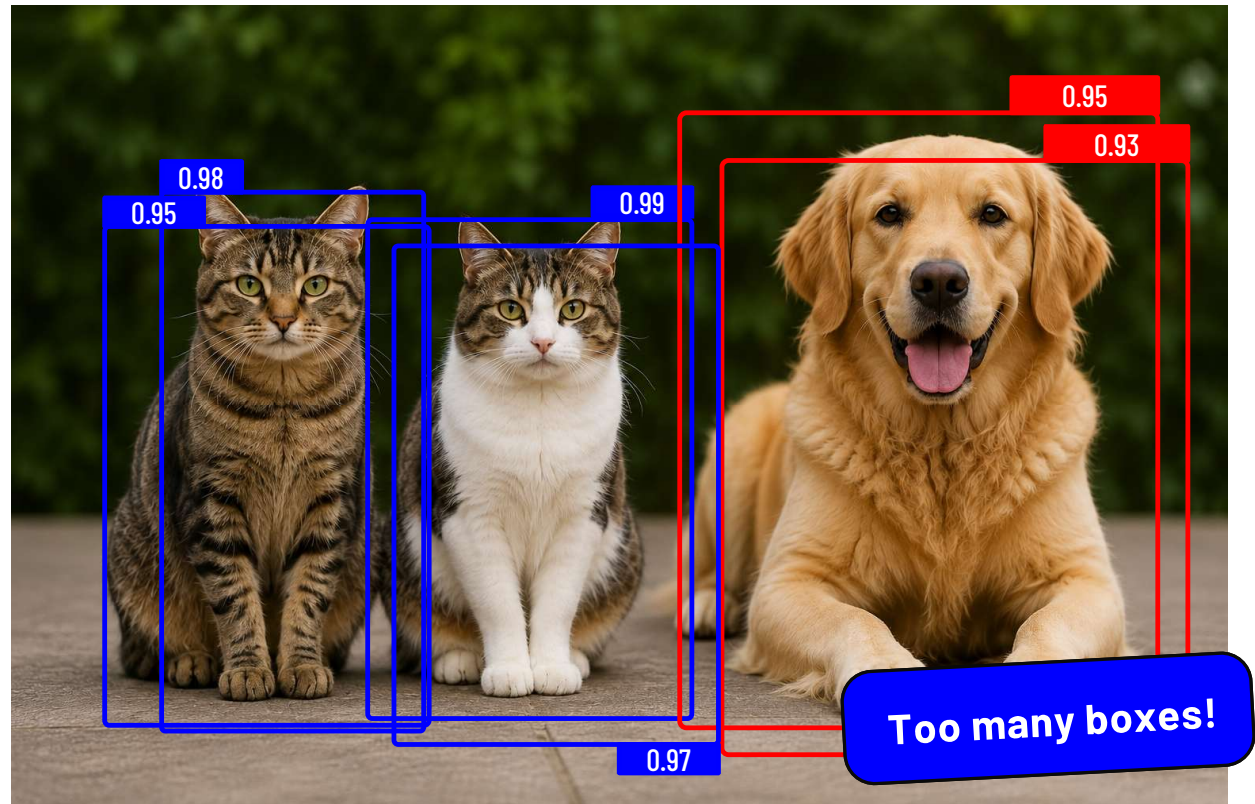


Image generated using ChatGPT. Bounding boxes added manually for demonstration purposes.

Non-Maximum Suppression

01

Sort

Take every detection and rank them **by confidence score**, highest first.

$\text{box}_1 \cdot 0.92$

$\text{box}_2 \cdot 0.88$

$\text{box}_3 \cdot 0.71$

$\text{box}_4 \cdot 0.54$

02

Pick

Pop the **highest-scoring** box off the top of the list. Move it to the keep-pile. This box is officially safe.



winner

03

Compare

For every remaining box, compute its **IoU** against the winner. How much do they overlap?

$\text{IoU}(\text{winner}, \text{box}_2) = 0.81$

$\text{IoU}(\text{winner}, \text{box}_3) = 0.62$

$\text{IoU}(\text{winner}, \text{box}_4) = 0.18$

04

Suppress

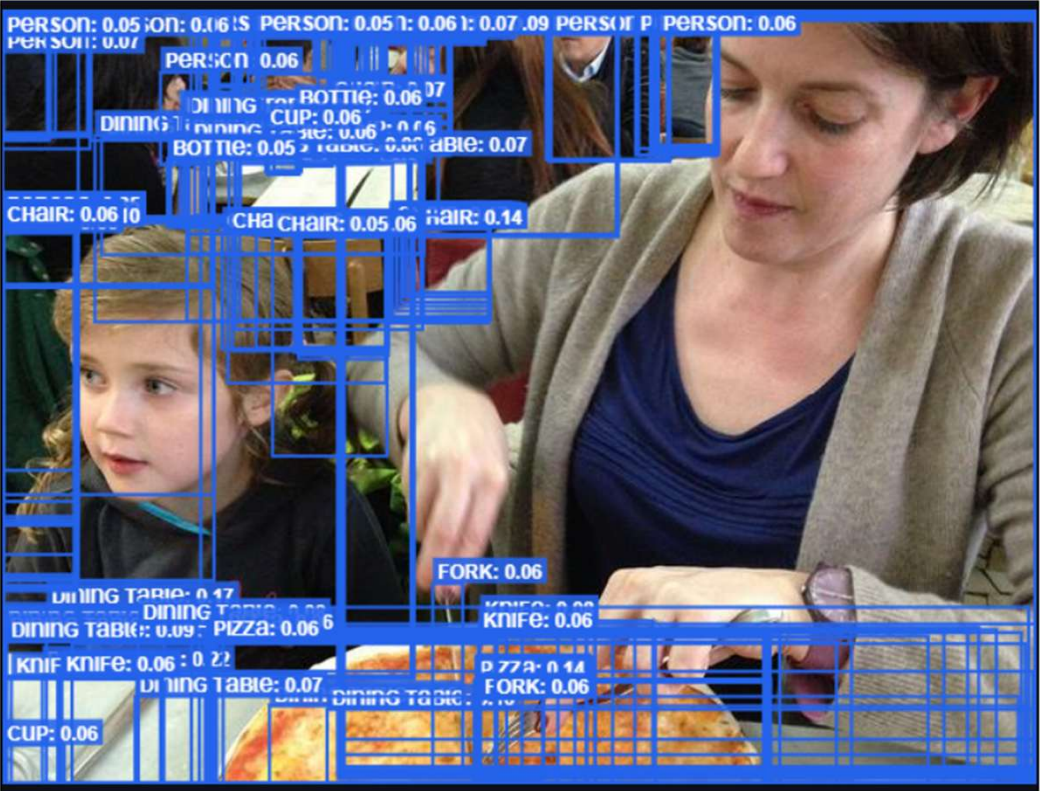
Drop any box whose IoU exceeds τ . Repeat from step 02 with what's left until the list is empty.

~~$\text{box}_2 \cdot 0.81$~~ X

~~$\text{box}_3 \cdot 0.62$~~ X

$\text{box}_4 \cdot 0.18$ ✓ stays

Extracts from Simulator - NMS



CANDIDATES (300)

ALL DETECTIONS · PRESS ► TO APPLY CONFIDENCE FILTER		
#0	person	0.91
#1	person	0.90
#2	person	0.89
#3	person	0.88
#4	person	0.88
#5	person	0.88
#6	person	0.88
#7	person	0.87
#8	person	0.87
#9	person	0.87
#10	person	0.86
#11	person	0.86

STEP 0 Confidence pre-filter applied.
180 box(es) with conf < 0.25 removed before NMS — see Discarded ↓

Extracts from Simulator - NMS

NON-MAXIMUM SUPPRESSION

Step: 1 / 17

Algorithm:

NMS

Soft-NMS

Compare

CONFIDENCE

0.25

IOU

0.50

IMAGE

STEP 0

Confidence pre-filter applied.

180 box(es) with conf < 0.25 removed before NMS — see Discarded ↓

CANDIDATES (120)

ACTIVE · CONF ≥ 0.25

#0	person	0.91
#1	person	0.90
#2	person	0.89
#3	person	0.88
#4	person	0.88
#5	person	0.88
#6	person	0.88
#7	person	0.87

DISCARDED

180 BOXES

IOU ≥ THR. (0)

CONF < 0.25 (180)

none yet

#120	person	0.25
#121	person	0.25
#122	person	0.24
#123	person	0.24
#124	chair	0.24
#125	person	0.24
#126	person	0.24
#127	chair	0.24

KEPT

✓ SELECTED BY NMS

Nothing kept yet.
Press ▶ to start.

LOG

START

Sorted 120 candidate(s) by confidence. Ready to run NMS.

Extracts from Simulator - NMS

NON-MAXIMUM SUPPRESSION Step: 2 / 17

Algorithm:

NMS

Soft-NMS

Compare

CONFIDENCE

0.25

 IOU

0.50

IMAGE

CANDIDATES (110)

IOU VS PIVOT #0 — THRESHOLD 0.50

#8	person	0.87	IoU 5%
#9	person	0.87	IoU 5%
#11	person	0.86	IoU 0%
#12	person	0.85	IoU 0%
#14	person	0.84	IoU 0%
#15	person	0.84	IoU 0%

X DISCARDED

IOU ≥ THR. (9)

CONF < 0.25 (100)

109 BOXES

#1	person	0.90	#120	person	0.25
#2	person	0.89	#121	person	0.25
#3	person	0.88	#122	person	0.24
#4	person	0.88	#123	person	0.24
#5	person	0.88	#124	chair	0.24
#6	person	0.88	#125	person	0.24
#7	person	0.87	#126	person	0.24
#10	person	0.86	#127	chair	0.24
#13	person	0.85	#128	chair	0.24

KEPT (1)

✓ SELECTED BY NMS

#0	person	0.91
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LOG

S1 PIVOT #0

threshold 0.50

Calculating IoU with 119 candidate(s) -- 9 suppressed

#1	100%	×
#2	99%	×
#3	99%	×
#4	99%	×
#5	100%	×
#6	99%	×
#7	99%	×
#8	5%	✓
#9	5%	✓
#10	99%	×
#11	0%	✓
#12	0%	✓
#13	99%	×
#16	5%	✓
#18	5%	✓
#20	0%	✓
#21	5%	✓
#23	0%	✓
#25	0%	✓
#26	5%	✓
#27	6%	✓
#30	6%	✓
#31	6%	✓
#32	6%	✓
#33	5%	✓

Extracts from Simulator - NMS

NON-MAXIMUM SUPPRESSION Step: 3 / 17

Algorithm:

NMS

Soft-NMS

Compare

CONFIDENCE

0.25

 IOU

0.50

IMAGE

CANDIDATES (100)

IOU VS PIVOT #8 — THRESHOLD 0.50

#11	person	0.86	IoU 9%
#12	person	0.85	IoU 9%
#14	person	0.84	IoU 3%
#15	person	0.84	IoU 3%
#17	person	0.82	IoU 3%
#19	person	0.82	IoU 3%

DISCARDED 198 BOXES

IOU ≥ THR. (10)	CONF < 0.25 (100)
#1 person 0.90	#120 person 0.25
#2 person 0.89	#121 person 0.25
#3 person 0.88	#122 person 0.24
#4 person 0.88	#123 person 0.24
#5 person 0.88	#124 chair 0.24
#6 person 0.88	#125 person 0.24
#7 person 0.87	#126 person 0.24
#10 person 0.86	#127 chair 0.24
#13 person 0.85	#128 chair 0.24

KEPT (2)

✓ SELECTED BY NMS

#0	person	0.91
#8	person	0.87

LOG

52 PIVOT #8 threshold 0.50

Calculating IoU with 109 candidate(s) — 9 suppressed

#9	99%	×
#11	9%	✓
#12	9%	✓
#14	3%	✓
#15	3%	✓
#16	98%	×
#17	3%	✓
#18	100%	×
#19	3%	✓
#20	9%	✓
#21	99%	×
#22	3%	✓
#23	9%	✓
#24	3%	✓
#25	9%	✓
#26	98%	×
#28	3%	✓
#29	3%	✓
#33	99%	×
#34	98%	×
#35	9%	✓
#36	3%	✓
#39	8%	✓
#41	9%	✓
#42	9%	✓

Extracts from Simulator - NMS

NON-MAXIMUM SUPPRESSION

Step: 4 / 17

Algorithm:

NMS

Soft-NMS

Compare

CONFIDENCE

0.25

IOU

0.50

IMAGE

CANDIDATES (89)

IOU VS PIVOT #11 — THRESHOLD 0.50

#14	person	0.84	IoU 19%
#15	person	0.84	IoU 18%
#17	person	0.82	IoU 19%
#19	person	0.82	IoU 19%
#22	person	0.82	IoU 18%
#24	person	0.81	IoU 19%

DISCARDED

208 BOXES

IOU ≥ THR. (28)		CONF < 0.25 (180)	
#1	person 0.90	#120	person 0.25
#2	person 0.89	#121	person 0.25
#3	person 0.88	#122	person 0.24
#4	person 0.88	#123	person 0.24
#5	person 0.88	#124	chair 0.24
#6	person 0.88	#125	person 0.24
#7	person 0.87	#126	person 0.24
#10	person 0.86	#127	chair 0.24
#13	person 0.85	#128	chair 0.24

KEPT (3)

✓ SELECTED BY NMS

#0	person	0.91
#8	person	0.87
#11	person	0.86

LOG

S3 PIVOT #11

threshold 0.50

Calculating IoU with 99 candidate(s) — 10 suppressed

#12	99%	×
#14	19%	✓
#15	18%	✓
#17	19%	✓
#19	19%	✓
#20	99%	×
#22	18%	✓
#23	99%	×
#24	19%	✓
#25	99%	×
#28	18%	✓
#29	18%	✓
#35	99%	×
#36	19%	✓
#39	98%	×
#41	99%	×
#42	99%	×
#43	18%	✓
#44	98%	×
#53	1%	✓
#56	1%	✓
#58	1%	✓
#59	1%	✓
#63	99%	×
#64	1%	✓

Extracts from Simulator - NMS

NON-MAXIMUM SUPPRESSION Step: 17 / 17

Algorithm:

NMS

Soft-NMS

Compare

CONFIDENCE

0.25

 IOU

0.50

IMAGE

CANDIDATES

IOU VS PIVOT #108 — THRESHOLD 0.50

DISCARDED

IOU ≥ THR. (104)	CONF < 0.25 (100)
#1 person 0.90	#120 person 0.25
#2 person 0.89	#121 person 0.25
#3 person 0.88	#122 person 0.24
#4 person 0.88	#123 person 0.24
#5 person 0.88	#124 chair 0.24
#6 person 0.88	#125 person 0.24
#7 person 0.87	#126 person 0.24
#10 person 0.86	#127 chair 0.24
#13 person 0.85	#128 chair 0.24

204 BOXES

KEPT (16)

SELECTED BY NMS

#0	person	0.91
#8	person	0.87
#11	person	0.86
#14	person	0.84
#27	person	0.81
#47	pizza	0.59
#48	pizza	0.53
#53	chair	0.50
#61	chair	0.45
#64	dining table	0.41
#67	person	0.40
#68	person	0.40
#69	person	0.40
#88	person	0.34
#104	chair	0.29
#108	person	0.28

LOG

S16 PIVOT #108 threshold 0.50

Calculating IoU with 1 candidate(s) — 1 suppressed

#119

99%

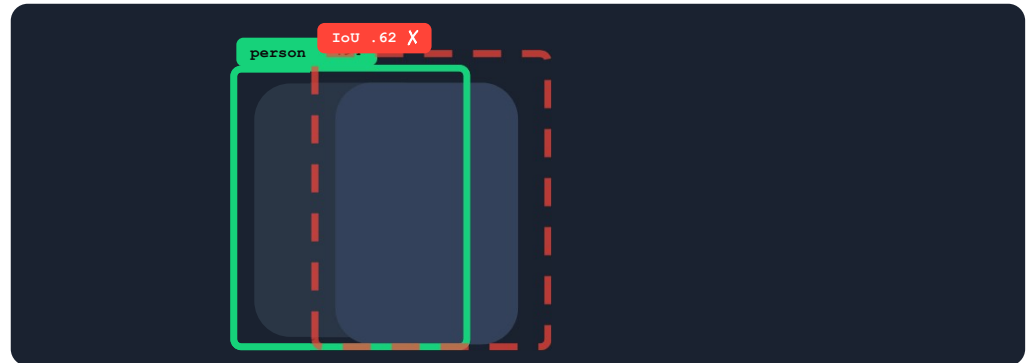
Don't Kill it. Quiet it

Hard NMS is binary - overlap above τ , the box is deleted forever. In crowded scenes that's catastrophic two pedestrians standing shoulder-to-shoulder share $\text{IoU} \approx 0.6$, and one of them **disappears** from the output.

Soft-NMS's - keep every box, but **scale down** its score by how much it overlaps the winner.

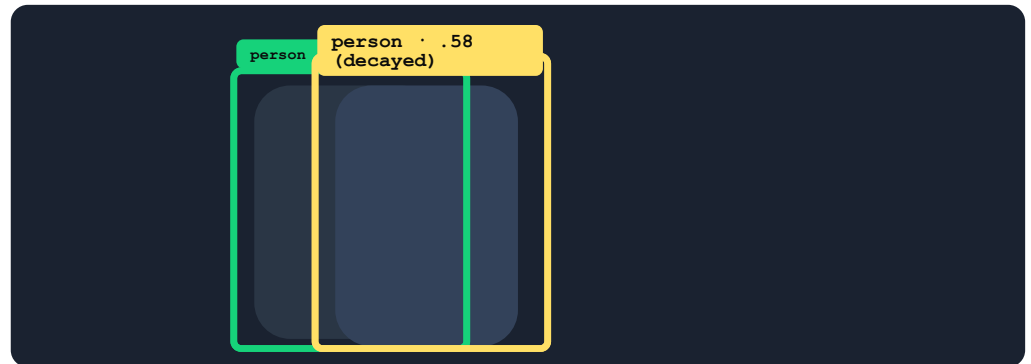
Hard NMS · $\tau=0.5$

person erased



Soft-NMS

both kept



Soft-NMS

Hard NMS is binary: any neighbor above the threshold is deleted outright. That's brutal when two different objects truly overlap, two pedestrians side-by-side, two cars at a junction.

Soft-NMS instead decays the score of each overlapping box by a function of IoU. Heavy overlaps shrink toward zero; light ones survive almost unchanged.

Final boxes are taken by score-thresholding after one pass - no boxes are ever deleted mid-loop.

Score decay functions

APPLY TO EVERY OVERLAPPING BOX

HARD NMS · REFERENCE

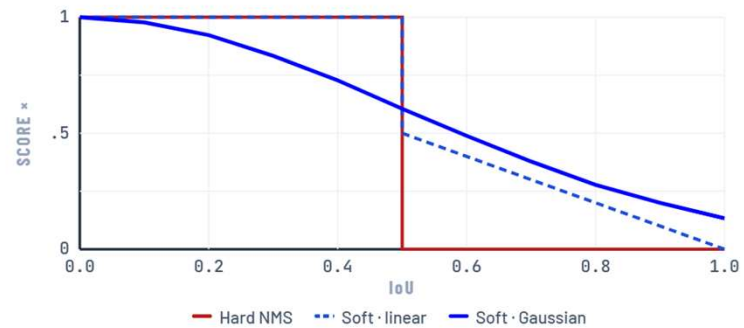
$$s_i = 0 \quad \text{if} \quad \text{IoU}(M, b_i) > N_t$$

SOFT-NMS · LINEAR

$$s_i = s_i \cdot (1 - \text{IoU}) \quad \text{if} \quad \text{IoU} > N_t$$

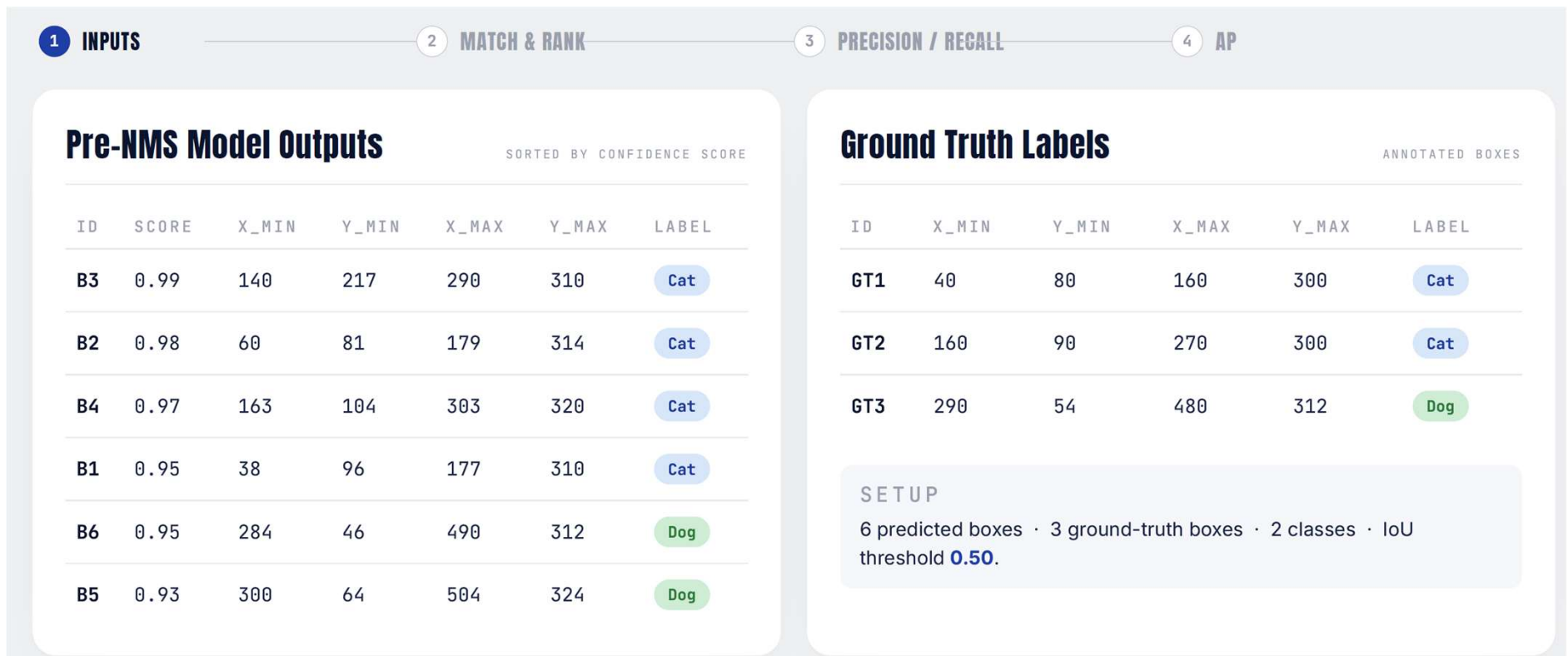
SOFT-NMS · GAUSSIAN

$$s_i = s_i \cdot \exp(-\text{IoU}^2 / \sigma) \quad \forall b_i$$

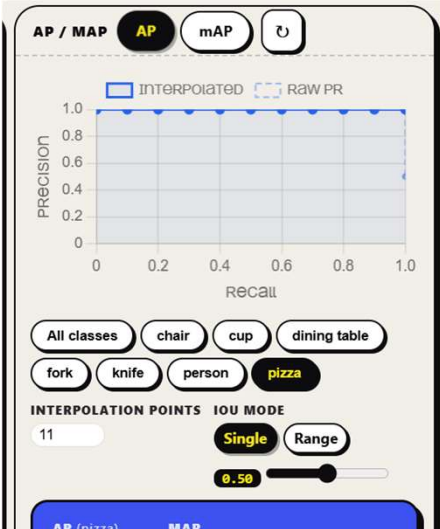
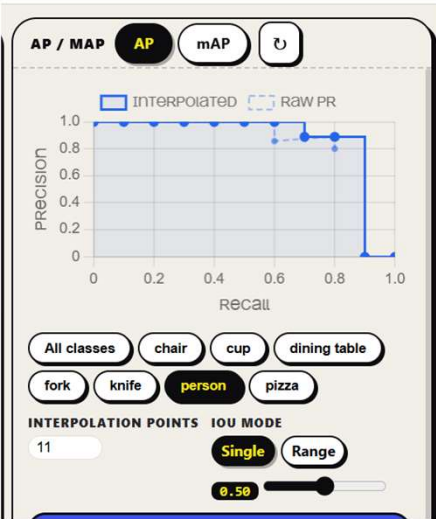
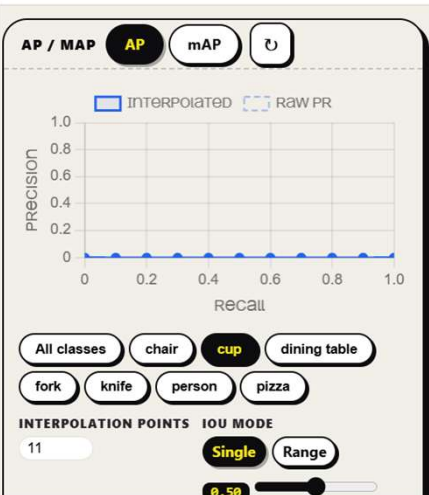
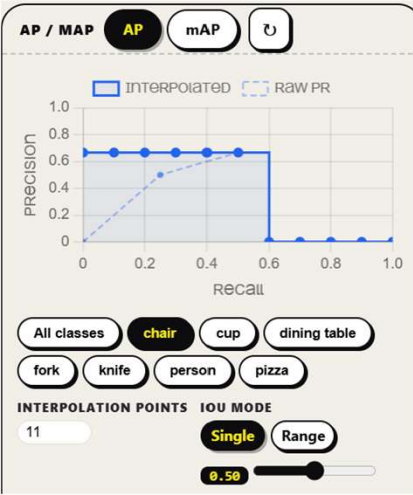


Average Precision & mean AP

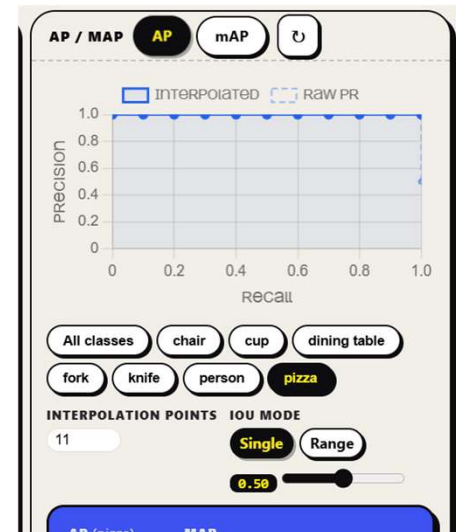
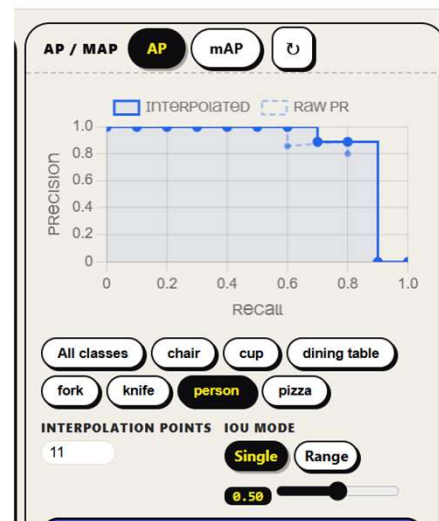
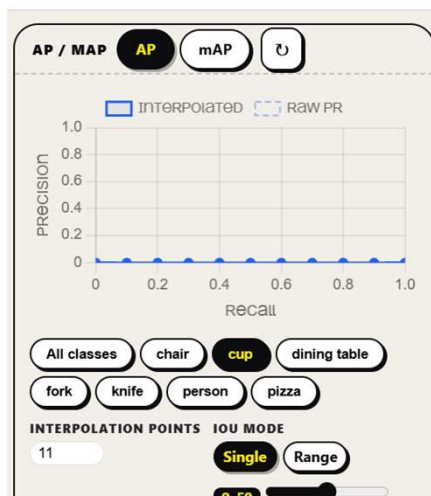
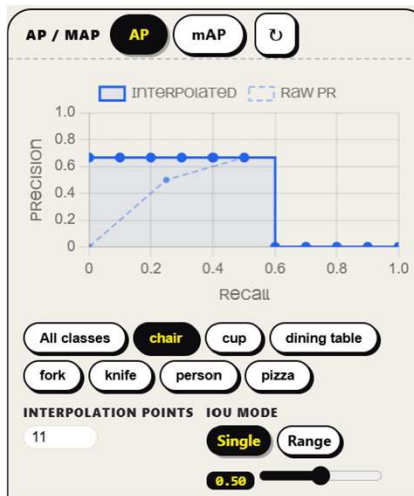
Average Precision



Extracts from Simulator - Average Precision



Extracts from Simulator – Average Precision



Case Study – Self Driving Cars

Self-driving stacks like **Waymo**, **Cruise**, and Tesla's Autopilot run a vision detector on every frame from every camera. A single sweep of the perception stack can emit **~25,000 anchor predictions** per camera long before any of it reaches the planner.

Without NMS, the planner would see forty boxes for every pedestrian and steer around all forty. NMS collapses that into one clean object track per agent fast enough to fit inside the 33 ms per-frame budget.

~25k

raw anchors / camera / frame

8

cameras around the vehicle

33 ms

budget per frame at 30 FPS

Thanks.

NMS

Keep one box per object.

IoU

The overlap metric.

Soft-NMS

Decay scores, don't drop.

P & R

Accuracy vs. coverage.

AP

Area under the PR curve.

mAP

Mean Average.